

Project Proposal

Identify dimensional structure of Visible and Near-Infrared (VNIR) + Short-wave Infrared (SWIR) Hyperspectral (HS) data. Data from study, <https://doi.org/10.3390/rs11080970>

Data collected from HySpex VNIR + SWIR sensors, giving a 6720x6427 region grid, 1mx1m spatial resolution (43 km² grid). Each pixel is captured in 430 bands of different wavelengths, range: 0.416–2.396 μm .

Most of these bands are highly correlated (e.g. adjacent wavelengths show similar reflectance) and can give redundant information, resulting in massive data files.

GOAL: Identify most impactful wavelengths/bands for future applications and research
HS sensors are very expensive, but crucial for modern systems supporting environmental conservation and wildfire risk assessment applications.

QUESTIONS:

- How many independent dimensions or features are actually needed to represent the data? Ex. 430 bands might effectively be represented in 10-20 dimensions.
- Which wavelengths change the most across the image, has the most variance?
- How much dimensionality reduction can be done to preserve spatially coherent vegetation structure, before spatial structure degrades?

METHODS:

Region-based sampling on dataset for specific methodology observation.

- Dense Vegetation, Grassland, Farm land, Bare ground/gravel

Select n samples, each n_i represent one value of the $p=430$ wavelengths

Construct data matrix $X \in \mathbb{R}^{n \times 430}$,

- Apply PCA via SVD
- Observe eigenvalue spectrum and explain variance
- Rank estimation
- Analyze PCA loadings vs wavelengths
- Employ K-means clustering to identify spectral groupings
- Visualize data in 2D/3D space

EXPECTED OUTCOMES:

Be able to identify intrinsic dimensionality of Hyperspectral data for vegetation observation, as well as the dominant spectral wavelengths

FUTURE RESEARCH:

Unsupervised fuel classification

Identify fuel moisture trends, for better informed prescribed burns practices

Sample Data

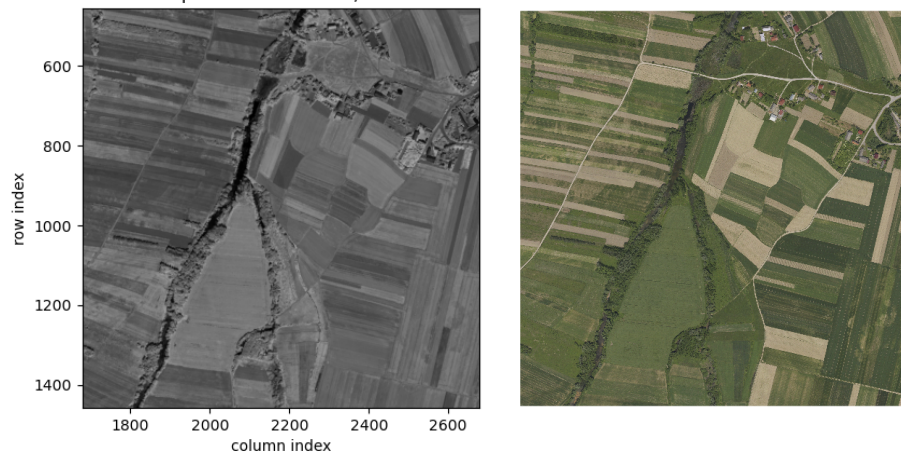
Dataset contains reflectance values, Here is sample table of values

Band p=100 values

	c1679	c1680	c1681	c1682	c1683	c1684	c1685	c1686	c1687	c1688
r457	0.295	0.269	0.278	0.292	0.299	0.298	0.284	0.284	0.283	0.281
r458	0.286	0.286	0.305	0.302	0.306	0.31	0.309	0.291	0.287	0.287
r459	0.299	0.293	0.3	0.3	0.304	0.305	0.301	0.297	0.3	0.295
r460	0.297	0.302	0.299	0.289	0.302	0.299	0.296	0.302	0.299	0.294
r461	0.307	0.308	0.302	0.303	0.285	0.294	0.296	0.295	0.292	0.296
r462	0.302	0.303	0.302	0.311	0.311	0.296	0.295	0.299	0.304	0.316
r463	0.26	0.292	0.285	0.297	0.289	0.269	0.277	0.288	0.295	0.309
r464	0.274	0.283	0.272	0.257	0.248	0.24	0.244	0.265	0.266	0.274
r465	0.273	0.275	0.274	0.278	0.274	0.268	0.266	0.261	0.261	0.249
r466	0.288	0.288	0.285	0.283	0.275	0.276	0.266	0.266	0.267	0.271

When values plotted on map: HS data on the left, RGB image taken from camera at same time on right. (RGB image just for reference, not going to be used as part of project)

Band p=100 at c0=1679, r0=457



Below is range of reflectance values

